Uniform spectral bounds for local Casimir sources

Abstract

We estimate the homogeneous four-variable polynomial of the full finite Casimir resolvent on the entire real spectral interval. Its decreasing matrix paths partition into a central sector and two one-sided sectors. For arbitrary face phases, we prove uniform exponential upper bounds in terms of an explicitly normalized triangle entropy. The central bound follows from exact cofactor interpolation at the physical nodes. The one-sided bound retains both unused spectral products and the signs inside an interval continuant; its two endpoint optimization has an explicit solution. The estimates include even and odd levels, the odd central-skipping term, and face variables fixed exactly at zero. They do not require a nonzero complex asymptotic or identify a volume.

1 The local bounds

The variables of a local face polynomial have a prescribed order: x, y are the outer variables, b the input middle variable, and c the output middle variable. Variables shared by different boxes must later be identified. This paper estimates one actual polynomial before that shared extraction. The estimates hold for all phases of the four variables, rather than only at their positive real values.

Fix an integer $N \geq 3$, and put

$$d = N - 1, \quad m = \lfloor N/2 \rfloor, \quad \theta = \pi/N, \quad q = e^{i\theta}, \quad Q = q^2.$$

The finite Fourier theorem in [1, Proposition 4.1 and equations (4.4)–(4.5)] identifies the physical pair Casimir Ω, one N-dimensional residue block at a time, with the polynomial operator specified in Section 2. The identification uses a finite Fourier transform, a transpose and a diagonal change of basis; it is stated precisely below. It also proves $\chi_N(\Omega) = 0$, where $\chi_N(z) = 2T_N(z/2) + 2$ and $T_N(\cos t) = \cos Nt$. The full resolvent numerator is the polynomial

$$V_N(\sigma; \Omega) = \frac{\chi_N(\sigma) - \chi_N(\Omega)}{\sigma - \Omega}.$$

The fraction is polynomial division, not an inverse at a spectral root. Its degree in σ is d; it retains the first derivatives at double roots as well as the values at distinct roots.

Write $H_V(x, b, c, y; \sigma)$ for its homogeneous face polynomial. Section 2 gives a division-free definition and fixes its sign. A decreasing path from row j to row i contributes the middle monomial $b^i c^{d-j}$. When $N = 2m$, its central sector has $i \leq m - 1, j \geq m$; when $N = 2m + 1$, it has $i \leq m, j \geq m$. The left sector has $j < m$; the right consists of the remaining paths above the central cut. Denote these three polynomials by $H_{V,\mathrm{C}}, H_{V,\mathrm{L}}, H_{V,\mathrm{R}}$. This is a partition of the full numerator paths, not a new spectral operator.

Set

$$\mathcal{B}(t) = -\int_0^t \log|2\sin v|\,dv, \qquad e(a) = \frac{\mathcal{B}(\pi a) + \mathcal{B}(\pi/2 - \pi a)}{\pi}, \quad 0 \le a \le \tfrac{1}{2}.$$

For nonnegative w_1, w_2, w_3, not all zero, define

$$\mathcal{Q}(w_1, w_2, w_3) = \max_{\substack{a_i \ge 0,\ \sum_i a_i = 1/2 \\ a_i = 0 \text{ if } w_i = 0}} \left\{ \sum_i e(a_i) + \sum_{w_i > 0} a_i \log w_i \right\}. \tag{1.1}$$

A variable fixed at zero has exactly zero load. It is never replaced silently by a small positive radius. The logarithmic integrals are finite at their endpoints. To fix their normalization, put $I = \int_0^{\pi/2} \log\sin t\,dt$. Reflection and the double-angle formula give $2I = I - (\pi/2)\log 2$. Hence $\mathcal{B}(0) = \mathcal{B}(\pi/2) = 0$, and $e(0) = e(1/2) = 0$.

In the theorem the lowercase letters denote nonnegative moduli. For $x + y > 0$, put

$$\begin{aligned} &Q_b = \mathcal{Q}(x, b, y), && Q_c = \mathcal{Q}(x, c, y), \quad X = Q_b + Q_c, \\ &A_{\mathrm{L}} = \max\{\log c, Q_b + \tfrac{1}{2}\log c\} && (c > 0), \\ &A_{\mathrm{R}} = \max\{\log b, Q_c + \tfrac{1}{2}\log b\} && (b > 0). \end{aligned} \tag{1.2}$$

If $c = 0$, the left polynomial is zero and its rate is interpreted as $-\infty$; the same convention applies on the right when $b = 0$.

Theorem 1.1 (Uniform local sector bounds)**.** *Fix nonnegative face moduli (x, b, c, y) with $x + y > 0$. For $s = \mathrm{C}, \mathrm{L}, \mathrm{R}$,*

$$\limsup_{N\to\infty} \frac{1}{N} \log \sup_{\substack{\sigma \in [-2,2] \\ |X_0| = x,\ |B_0| = b,\ |C_0| = c,\ |Y_0| = y}} |H_{V,s}(X_0, B_0, C_0, Y_0; \sigma)| \le \begin{cases} X & s = \mathrm{C}, \\ A_{\mathrm{L}} & s = \mathrm{L}, \\ A_{\mathrm{R}} & s = \mathrm{R}. \end{cases} \tag{1.3}$$

The limit superior is over all integer levels, including both parities and composite levels. The statement is uniform when the zero pattern is fixed and every nonzero modulus varies in a fixed compact subset of $(0, \infty)$. Thus, for every $\varepsilon > 0$, there is one threshold N_0 on such a compact set for which the indicated rate plus ε bounds the normalized logarithm at every $N \ge N_0$.

The supremum precedes the limit, so sampled spectral zeros are included. The proof never divides a finite numerator by such a zero, and never replaces a negative tail coefficient by its absolute value inside the recurrence.

The admissible vertex with load $1/2$ on b gives

$$\mathcal{Q}(x, b, y) \ge \tfrac{1}{2}\log b \quad (b > 0), \qquad \max\{X, A_{\mathrm{L}}, A_{\mathrm{R}}\} = \max\{X, \log b, \log c\}, \tag{1.4}$$

with undefined logarithms omitted. Each mixed expression in (1.2) is at most X. This identifies the full local spectator bound, but does not identify an optimized collection of local bounds with a shared coefficient.

2 The exact polynomial and its path partition

We import only the finite operator identification from [1]. The matrix needed in the estimates is explicit. On $1, z, \dots, z^d$, let

$$\mathsf{C}z^j = 2\cos((2j+1)\theta)z^j - s_j^2(x+y)z^{j-1} - s_j s_{j-1}xy\, z^{j-2}, \qquad s_j = 2\sin(j\theta). \tag{2.1}$$

Negative powers vanish because $s_0 = 0$. In the finite Fourier theorem $\mathsf{C} = -T$. To specify the imported identification, let Ω_s be the block of the physical pair operator on

$$e_u \otimes e_{[s-u]_N}, \qquad 0 \le u < N, \qquad 0 \le s < N.$$

Here $e_0, \dots, e_d$ is its standard one-strand basis, and $[v]_N$ is the representative in $\{0, \dots, N-1\}$. For $y \neq 0$, specialize $x = Q^{s+1}y$, set $\zeta_u = -q^{-1}xQ^{-u}$, and define $\mathsf{E}_{uj} = \zeta_u^j$, $G_{uu} = q^{du}$. Then

$$\mathsf{E}\mathsf{C}\mathsf{E}^{-1} = G^{-1}\Omega_s^{\mathsf{T}}G. \tag{2.2}$$

The N blocks form the N^2-dimensional pair space. The identity includes their nilpotent parts, not just eigenvalues. The face polynomial retains independent x, y; its specialized values recover the matrix entries by the adapter in [1, equation (4.5)]. If $f(\mathsf{C})_{ij}$ is the coefficient from z^j to z^i, the homogeneous polynomial is

$$H_f(x, b, c, y) = \sum_{0 \le i \le j \le d} f(\mathsf{C})_{ij} b^i c^{d-j}.$$

Its entries are polynomials in x, y, and its total face degree is d.

Set

$$\sigma = -2 - u, \qquad a_j = 2 + 2\cos((2j+1)\theta), \qquad \alpha_j = a_j + u, \qquad A = x + y, \quad P = xy. \tag{2.3}$$

The triangular matrix $\mathsf{M} = 2I + \mathsf{C} + uI$ has diagonal α_j, and lowering entries $-s_j^2 A$, $-s_j s_{j-1} P$. Its diagonal product is $(-1)^N \chi_N(-2-u)$. Consequently

$$\mathcal{P}_N(x, b, c, y; u) := (-1)^d H_V(x, b, c, y; -2-u) = \sum_{i \le j} (\operatorname{adj} \mathsf{M})_{ij} b^i c^{d-j}. \tag{2.4}$$

The identity follows first where M is invertible and then by polynomial equality, including every singular spectral value.

Expand its triangular inverse by the strictly lowering part and multiply by all diagonal factors. A path $j \to i$ uses steps of length one or two. Its weight is the product of $s_r^2 A$ for a step $r \to r-1$, $s_r s_{r-1} P$ for a step $r \to r-2$, and α_r for precisely the rows it does not visit. The middle monomial $b^i c^{d-j}$ completes its term. This proves the path rule, including the case of a zero unused factor.

Partition $\mathcal{P}_N = \mathcal{P}_{\mathrm{C}} + \mathcal{P}_{\mathrm{L}} + \mathcal{P}_{\mathrm{R}}$ by the sectors in the introduction. Their common sign relative to H_V does not affect modulus bounds. Every sector has spectral degree at most d. At $u = 0$, every path weight is nonnegative. At odd N, a path missing the middle row has its unused factor $a_m = 0$.

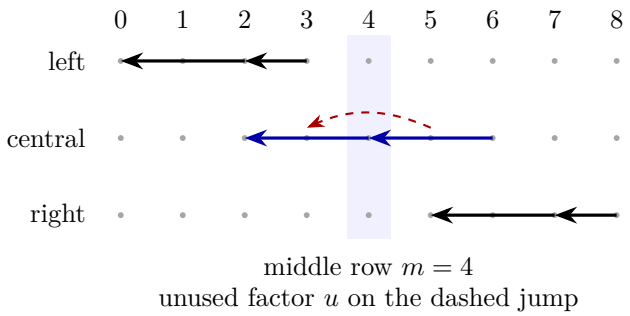


Figure 1: The partition at the illustrative odd level $N = 9$. A central path may visit row m, or skip it by $m + 1 \to m - 1$. The skipped row contributes its unused factor u, retained in the exact central polynomial even though it vanishes at $u = 0$.

2.1 Cleared cofactors

Let $R_j(b; u)$ be the determinant obtained from the block of M on rows $0, \dots, j$ by replacing its last row with $(1, b, \dots, b^j)$. Its path expansion gives

$$\begin{aligned} R_0 = 1, \qquad R_{-1} = R_{-2} = 0, \\ R_j(b; u) = b^j \prod_{r<j} \alpha_r + s_j^2 A R_{j-1}(b; u) + s_j s_{j-1} \alpha_{j-1} P R_{j-2}(b; u). \end{aligned} \tag{2.5}$$

The first term at $j = 0$ is the empty product one. Each R_j has face degree j and spectral degree at most j. Put

$$D_b = R_{m-1}(b; u), \qquad E_b = \alpha_{m-1} R_{m-2}(b; u),$$

with $E_b = 0$ when $m = 1$. Subscripts b, c here denote substitution of the middle variable, not differentiation.

Lemma 2.1 (Exact central factors)**.** *At $N = 2m$,*

$$\mathcal{P}_{\mathrm{C}} = s_m^2 A D_b D_c + s_m s_{m-1} P (D_b E_c + E_b D_c). \tag{2.6}$$

At $N = 2m + 1$, with $G_b = R_m(b; u)$,

$$\mathcal{P}_{\mathrm{C}} = G_b G_c + u s_m^2 P D_b D_c. \tag{2.7}$$

Both are polynomial identities for every u.

Proof. An even central path crosses the cut exactly once, by $m \to m - 1$, $m + 1 \to m - 1$, or $m \to m - 2$. Their weights give the three terms. Reflection $r \mapsto d - r$ identifies upper and lower cofactors: $a_{d-r} = a_r$, and the reflected step products agree. At odd level, paths visiting m split there to give $G_b G_c$. A path skipping it has the unique jump $m + 1 \to m - 1$, with weight $s_{m+1} s_m P = s_m^2 P$; its unused diagonal is u. This proves the second term without dividing by u. □

The same reflection interchanges the left and right sectors and b, c. A one-step at r reflects to one at $N - r$, with $s_{N-r} = s_r$; a two-step reflects to one at $N + 1 - r$, with product $s_{r-1} s_r$. It therefore suffices to estimate the left sector.

3 Exact nodal control of the central sector

For face polynomials, $F \preccurlyeq G$ means that every coefficient of $G - F$ is nonnegative real. The symbol $|F|_{\mathrm{coef}}$ replaces each coefficient by its modulus. Put $J = \lfloor (N-1)/2 \rfloor$. The $J+1$ distinct physical nodes in the u coordinate are

$$u_a = -a_a, \qquad 0 \le a \le J.$$

They include zero at odd level, but not at even level.

Lemma 3.1 (Nodal positivity and comparison)**.** *For every $0 \le j, a \le J$,*

$$0 \preccurlyeq R_j(b; u_a) \preccurlyeq R_j(b; 0).$$

Proof. If $j \le a$, every diagonal factor used in (2.5) lies between zero and its value at $u = 0$; induction proves the claim. If $j > a$, the inhomogeneous term vanishes after row a. Every surviving path must visit row a, so

$$R_j(b; u_a) = K_{a,j}(u_a) R_a(b; u_a),$$

where the interval continuant satisfies

$$K_{i,i} = 1, \quad K_{i,i-1} = 0, \quad K_{i,j} = s_j^2 A K_{i,j-1} + s_j s_{j-1} \alpha_{j-1} P K_{i,j-2}.$$

Normalize $q_r = K_{a,a+r} / \prod_{v=a+1}^{a+r} s_v^2$. For $r \ge 2$, $j = a + r \le J$, product-to-sum gives

$$a_{j-1} - a_a = -s_{j+a} s_{j-a-1}, \qquad s_j s_{j-1} - s_{j+a} s_{j-a-1} = s_a s_{a+1}.$$

All the indicated sines are nonnegative. Thus

$$q_0 = 1, \quad q_1 = x + y, \quad q_r = (x+y) q_{r-1} - \gamma_r x y q_{r-2}, \qquad \gamma_r = 1 - \frac{s_a s_{a+1}}{s_j s_{j-1}} \in [0, 1]. \tag{3.1}$$

The identity

$$q_r - x q_{r-1} = y(q_{r-1} - x q_{r-2}) + (1 - \gamma_r) x y q_{r-2}$$

proves simultaneous coefficientwise positivity, starting from $q_0 = 1, q_1 - x q_0 = y$. Also $q_r \preccurlyeq (x+y)^r$. At $u = 0$, the interval continuant contains this all-one-step term and other nonnegative terms. Hence $K_{a,j}(u_a) \preccurlyeq K_{a,j}(0)$. The product $K_{a,j}(0) R_a(b; 0)$ is the subset of paths in $R_j(b; 0)$ whose start is at or below a and which visit a. This proves the assertion without assuming positivity of the signed recurrence. □

Lemma 3.2 (Exact interpolation bound)**.** *An absolute constant C gives, for $j \le J$,*

$$|R_j(b; u)|_{\mathrm{coef}} \preccurlyeq C(1 + \log N) R_j(b; 0), \qquad -4 \le u \le 0. \tag{3.2}$$

Proof. Each face coefficient is a polynomial in u of degree at most $j \le J$. This interpolation is not applied to the degree-d full resolvent. Use $z = -1 - u/2$ and $\phi_k = (2k+1)\pi/N$, $0 \le k < N$. For a polynomial $p(z)$ of degree at most J, discrete Fourier inversion gives

$$p(\cos t) = \frac{1}{N} \sum_{k=0}^{N-1} D_J(t - \phi_k) p(\cos \phi_k), \qquad D_J(v) = \sum_{\ell=-J}^{J} e^{i\ell v}.$$

The difference of any two retained modes has magnitude less than N, so its grid average is a Kronecker delta. Pairing equal cosine nodes gives ordinary interpolation, with the unpaired point -1 at odd N. Since

$$|D_J(v)| \le \min(2J+1, |\sin(v/2)|^{-1}),$$

the normalized sum of absolute kernel values on any translate of the grid is at most $C(1+\log N)$. The two nearest points cost a constant; successive distances k/N cost at most C/k. Pairing cannot increase this bound. Apply it to each coefficient and use Lemma 3.1. □

Proposition 3.3 (Central polynomial comparison)**.** *For an absolute C and every $N \ge 3$,*

$$|\mathcal{P}_{\mathrm{C}}(u)|_{\mathrm{coef}} \preccurlyeq C(1+\log N)^2 \mathcal{P}_{\mathrm{C}}(0), \qquad -4 \le u \le 0. \tag{3.3}$$

Proof. Apply (3.2) to the exact central factors. The scalar factors $|u|, |a_{m-1}+u|$ are at most four. At even $N \ge 4$,

$$D_b(0) \succcurlyeq s_{m-1}^2 (x+y) R_{m-2}(b;0), \qquad (x+y)^2 \succcurlyeq 2xy.$$

The sine is uniformly bounded below. Thus either mixed term $xyD_b(0)R_{m-2}(c;0)$ is bounded coefficientwise by a constant times $(x+y)D_b(0)D_c(0)$, already present at zero. At odd levels,

$$G_b(0) \succcurlyeq s_m^2 (x+y) D_b(0), \qquad G_b(0)G_c(0) \succcurlyeq 2s_m^4 xy D_b(0) D_c(0).$$

Here s_m is bounded below, including $N = 3$. The skipped-middle term is therefore controlled as well. □

4 A signed interval transfer estimate

This section bounds the signed recurrence. It is not a Perron estimate for its entrywise absolute value. Parametrize the real interval by

$$\sigma = -2\cos(2\pi\delta), \qquad u = -4\sin^2(\pi\delta), \qquad 0 \le \delta \le \tfrac{1}{2}.$$

For positive outer sum $A = x+y$ and $P = xy \ge 0$, define

$$\begin{aligned} a_\delta(t) &= 4(\cos^2(\pi t) - \sin^2(\pi\delta)), \\ \rho_\delta(t) &= A\sin(\pi t) + \sqrt{(x-y)^2\sin^2(\pi t) + 4P\cos^2(\pi\delta)}, \\ \lambda_\delta(t) &= 2\sin(\pi t)\rho_\delta(t). \end{aligned} \tag{4.1}$$

On each compact set of allowed moduli,

$$ct \le \rho_\delta(t) \le C, \qquad ct^2 \le \lambda_\delta(t) \le C, \qquad 0 < t \le \tfrac{1}{2}, \tag{4.2}$$

uniformly in δ.

Lemma 4.1 (Uniform signed transfer)**.** *For every $\varepsilon > 0$, all sufficiently large N satisfy*

$$|K_{i,j}| \le \exp\left\{ N \int_{i/N}^{j/N} \log \lambda_\delta(t)\, dt + \varepsilon N \right\}, \qquad 0 \le i \le j \le m. \tag{4.3}$$

The threshold is uniform in endpoints, real spectral parameter, and each fixed compact set of allowed moduli.

Proof. Normalize $L_{i,j} = K_{i,j}/\prod_{r=i+1}^{j} s_r$. Its step matrix is

$$T_h(t,\delta) = \begin{pmatrix} 2A\sin(\pi t) & 4P[\cos^2(\pi(t-h/2)) - \sin^2(\pi\delta)] \\ 1 & 0 \end{pmatrix}, \quad h = N^{-1}, \quad t = jh.$$

The limiting matrix has real eigenvalues

$$A\sin(\pi t) \ \pm \ \sqrt{(x-y)^2\sin^2(\pi t) + 4P\cos^2(\pi\delta)}.$$

Its spectral radius is $\rho_\delta(t)$, including coincident roots. Use the Euclidean operator norm. Fix $\eta > 0$ and restrict to $t \in [\eta, 1/2]$, where $\rho_\delta(t) \geq c_\eta > 0$. Cayley–Hamilton gives uniformly

$$\|T^k\| \leq C_\eta(k+1)\rho(T)^k. \tag{4.4}$$

For distinct eigenvalues the power divided difference is a sum of k terms, each bounded by ρ^{k-1}. The polynomial identity extends to coincident eigenvalues by its derivative value. Bounded entries and the positive lower bound on ρ give C_η. No eigenbasis or its condition number is used.

Choose a fixed block length k for which $\log(2C_\eta(k+1))/k$ is smaller than a prescribed error. Within a block, the exact matrices differ from its first limiting matrix by $O_k(N^{-1})$. Expanding the finite product one factor at a time bounds its difference from T^k by the same order. For large N this is at most $C_\eta(k+1)\rho^k$. Multiplying block bounds and using uniform continuity of $\log\rho$ gives its integral plus the prescribed error. Incomplete blocks have at most $2k$ factors and cost $o(N)$, uniformly in the endpoints.

For $0 < t < \eta$, bounded entries cost at most $C^{\eta N + O(1)}$. The omitted logarithmic integral has absolute value at most $C\eta\log(2/\eta)$, by (4.2). The exact scalar factors s_r have the same integrable endpoint bound, and their logarithmic Riemann sums converge uniformly over subintervals. Choose η first, then the fixed block length, and finally N. Restoring these factors proves the estimate. If a perturbation of the reference matrices is wanted, it must be chosen after the block length; a sufficiently small fixed perturbation then fits the same finite-product error. No complex perturbation is needed for the theorem here. □

4.1 Unused spectral rows

The exact diagonals are $\alpha_r = a_\delta((r+1/2)/N)$. Uniformly in δ and interval endpoints,

$$\sum_{r=l}^{k-1} \log|\alpha_r| \leq N\int_{l/N}^{k/N} \log|a_\delta(t)|\,dt + o(N). \tag{4.5}$$

A sampled zero makes the left side $-\infty$, which causes no upper-bound problem. This convention describes a zero product; no finite division by it is made.

For a proof, truncate the logarithm from below at $-M$. The truncated function is jointly continuous, so its Riemann sums converge uniformly. The factorization

$$a_\delta(t) = 4\cos(\pi(t+\delta))\cos(\pi(t-\delta))$$

shows that $\{|a_\delta| < r\}$ has length at most $C\sqrt{r}$, uniformly in δ. Integrating this distribution bounds the omitted logarithmic tails uniformly by a quantity tending to zero as $M \to \infty$. The argument applies to every subinterval.

There is also the exact complete-prefix identity

$$\int_0^{1/2} \log|a_\delta(t)|\,dt = 0. \tag{4.6}$$

Shift the two cosine integrals and use the reflection and period of $\log|2\cos(\pi t)|$. Together they have the integral of a full period, which is zero. Integrability includes $\delta = 0, 1/2$. We use this identity only after both actual unused products have been bounded separately.

5 The normalized triangle action

For $b > 0$, define

$$F_b(\mu,\delta) = \mu\log b + \int_0^{\mu} \log|a_\delta(t)|\,dt + \int_{\mu}^{1/2} \log\lambda_\delta(t)\,dt. \tag{5.1}$$

When $b = 0$, only $\mu = 0$ is allowed. The exact start decomposition is

$$R_n(b;u) = \sum_{i=0}^{n} b^i \left(\prod_{r<i}\alpha_r\right) K_{i,n}.$$

Lemma 4.1, (4.5), and at most $N+1$ starts give, for $n = N/2 + O(1)$ and $n \le m$,

$$|R_n(b;u)| \le \exp\left\{N\left[\max_{0\le\mu\le 1/2} F_b(\mu,\delta) + \varepsilon\right]\right\} \tag{5.2}$$

uniformly for large N. Replacing n/N by $1/2$ costs $o(N)$, since λ_δ is uniformly positive at $1/2$.

For $0 \le \delta < 1/2$, put $s = \sin(\pi\delta)$, $k = \cos(\pi\delta)$, and

$$T_b = \sqrt{bA + P + b^2 s^2}, \qquad \mathcal{D} = A^2 - 4Ps^2, \qquad \mathcal{E}_b = A + 2bs^2 + 2sT_b.$$

These scalar symbols are distinct from the finite cofactors D_b, E_b. The stationary point before the sign change $t_c = 1/2 - \delta$ is

$$\mu_b = \frac{1}{\pi}\arctan\frac{bk}{T_b}, \qquad \sin^2(\pi\mu_b) = \frac{k^2 b^2}{(b+x)(b+y)}. \tag{5.3}$$

For $b = 0$ set $\mu_b = 0$, and write $Q_b(\delta) = F_b(\mu_b,\delta)$.

Lemma 5.1 (Calibration and spectral derivative)**.** *For the allowed nonnegative weights,*

$$Q_b(0) = \mathcal{Q}(x,b,y). \tag{5.4}$$

For $0 < \delta < 1/2$,

$$Q_b'(\delta) = -\log(\mathcal{E}_b/\sqrt{\mathcal{D}}) \le 0. \tag{5.5}$$

If $b = P = 0$, *the action is the constant* $\frac{1}{2}\log A$. *The endpoint and zero-weight cases follow from the exact sole-start or one-step formulas and their continuous limits.*

Proof. The stationary equation is $b\,a_\delta(\mu) = \lambda_\delta(\mu)$ where $a_\delta > 0$. Squaring its positive sides gives (5.3).

First fix the constant in (5.4). For $0 < a < 1/2$,

$$e'(a) = \log\cot(\pi a), \qquad e''(a) = -\frac{\pi}{\sin(\pi a)\cos(\pi a)} < 0.$$

At positive weights the entropy maximizer is unique and has loads

$$a_w = \frac{1}{\pi}\arctan\frac{w}{\sqrt{xb + xy + by}}, \qquad w = x, b, y.$$

The tangent addition identity makes their sum $1/2$, and the three constrained derivatives are equal. Hence the derivative of $\mathcal{Q}(x, b, y)$ with respect to $\log b$ is $a_b = \mu_b|_{\delta=0}$. The envelope rule gives the same derivative for $F_b(\mu_b, 0)$, so their difference is independent of $b > 0$. As $b \to \infty$, both minus $\frac{1}{2}\log b$ tend to zero. For the entropy this follows also directly from the displayed loads: the other loads and their entropy terms tend to zero, and their products with $\log b$ vanish. For the integral, $1/2 - \mu_b = O(b^{-1/2})$, the full prefix integral is zero, and the remaining logarithmic tails vanish. This proves equality without a geometric normalization. Zero weights follow by continuity on the constrained faces of the load simplex and by the dominated logarithmic integrals.

For the derivative, $\partial_{\log b} Q_b = \mu_b$, and direct differentiation gives

$$\partial_\delta \mu_b = -\frac{bs}{T_b}, \qquad \partial_{\log b}\log(\mathcal{E}_b/\sqrt{\mathcal{D}}) = \frac{bs}{T_b}.$$

Thus $Q_b' + \log(\mathcal{E}_b/\sqrt{\mathcal{D}})$ is independent of b. Determine it at $b = 0$, retaining the integration constant. For $P > 0$, set $\lambda = 2\sqrt{P}/A \in (0, 1]$. Differentiating the sole-start integral and putting $\vartheta = \pi t$ gives

$$Q_0' = -\lambda^2 sk \int_0^{\pi/2} \frac{d\vartheta}{W(\vartheta)(\sin\vartheta + W(\vartheta))}, \quad W(\vartheta) = \sqrt{(1-\lambda^2)\sin^2\vartheta + \lambda^2 k^2}.$$

Rationalization and $z = \cos\vartheta$ turn this into

$$-sk\int_0^1 \left\{\frac{1}{\sqrt{1-z^2}} - \frac{1}{\sqrt{1-\lambda^2 s^2 - (1-\lambda^2)z^2}}\right\}\frac{dz}{z^2 - s^2}.$$

The apparent pole at $z = s$ is removable. For either term, $B - Cs^2 = k^2$, and differentiation verifies

$$\int \frac{dz}{(z^2 - s^2)\sqrt{B - Cz^2}} = \frac{1}{2sk}\log\left|\frac{kz/\sqrt{B - Cz^2} - s}{kz/\sqrt{B - Cz^2} + s}\right|.$$

Subtract the antiderivatives before evaluating. Their singularities at $z = s$ cancel. At zero both are zero; at one the first tends to zero and the second gives the finite remaining term. The result is

$$Q_0' = -\operatorname{artanh}(\lambda s) = -\log\frac{A + 2s\sqrt{P}}{\sqrt{A^2 - 4Ps^2}}.$$

This proves (5.5) for all $b \geq 0$. When $P = 0$, the exact tail root is $4A\sin^2(\pi t)$; direct differentiation gives the same formula. It also follows by dominated convergence as $P \downarrow 0$. If $b = P = 0$, the integral is $\frac{1}{2}\log A$. Finally $\mathcal{E}_b \geq \sqrt{\mathcal{D}} > 0$ in the open interval. The possible singularity at $x = y, \delta = 1/2$ is logarithmic and integrable. □

Lemma 5.2 (The cofactor maximum)**.** *For $b > 0$,*

$$\max_{0\le\mu\le 1/2} F_b(\mu,\delta) = \max\{Q_b(\delta), \tfrac{1}{2}\log b\}.$$

For $b = 0$, the sole value is $Q_0(\delta)$.

Proof. At a regular point, $\partial_\mu F_b = \log(b|a_\delta|/\lambda_\delta)$. Writing $w = k^2/\sin^2(\pi\mu)$, the ratio is

$$\frac{2b|w-1|}{A+\sqrt{(x-y)^2+4Pw}}.$$

Before t_c, w decreases from infinity to one and this ratio strictly decreases from infinity to zero. To check its monotonicity for $w > 1$, the positive derivative numerator after cancellation is

$$A + \frac{(x-y)^2+2Pw+2P}{\sqrt{(x-y)^2+4Pw}}.$$

The case $P = 0$ is immediate. Thus the unique maximum there is μ_b. After t_c the ratio increases with μ, so every critical point is a minimum. The last endpoint has value $\frac{1}{2}\log b$ by (4.6); the first is below the pre-sign-change maximum. At $\delta = 1/2$, the ratio is the constant $b/\max(x,y)$, and the endpoint values prove the assertion. The continuous limit is $Q_b(1/2) = \frac{1}{2}\log\max(x,y)$. □

At $\delta = 0$, these lemmas and $\mathcal{Q}(x,b,y) \ge \frac{1}{2}\log b$ imply

$$R_{N/2+O(1)}(b;0) \le \exp\{N(\mathcal{Q}(x,b,y)+\varepsilon)\}.$$

Use this in the exact central factorization at zero. The scalar coefficients and face monomials are bounded on each fixed compact. Proposition 3.3 then proves the central part of Theorem 1.1, including arbitrary face phases.

6 The one-sided envelope with both unused products

For $0 \le i \le j < m$, the exact left summand is

$$A_{\mathrm{rest}}(u) b^i c^{d-j} \left(\prod_{r<i}\alpha_r\right)\left(\prod_{j<r<m}\alpha_r\right) K_{i,j}(u), \tag{6.1}$$

where $A_{\mathrm{rest}} = \prod_{r=m}^{d}\alpha_r$ is the whole omitted upper-half product. On the real interval it has no positive exponential cost. At $N = 2m$,

$$A_{\mathrm{rest}}(u) = 2(-1)^m T_m(\sigma/2), \qquad |A_{\mathrm{rest}}| \le 2.$$

Its roots and leading coefficient prove the identity. At $N = 2m+1$, it is, up to sign, $(\sigma+2)P_m(\sigma)$, where

$$P_m(2\cos t) = \frac{\cos((m+1/2)t)}{\cos(t/2)}.$$

The quotient is a finite alternating sum of m paired exponentials and one constant, so $|P_m| \le 2m+1 = N$. Hence $|A_{\mathrm{rest}}| \le 4N$, including the removable endpoint. These are the original unused products, not auxiliary normalizations.

For $b, c > 0$, apply the transfer and Riemann estimates separately to the two outside products in (6.1). Only then use (4.6). With $\mu = i/N, \nu = j/N$, the upper action becomes

$$\begin{aligned} F_{\mathrm{L}}(\mu,\nu,\delta) &= \mu \log b + (1-\nu)\log c + \int_{\mu}^{\nu} \log L_\delta(t)\, dt, \\ L_\delta(t) &= \frac{\lambda_\delta(t)}{|a_\delta(t)|}, \qquad 0 \le \mu \le \nu \le \tfrac{1}{2}. \end{aligned} \tag{6.2}$$

This quotient is only continuum notation. No finite division by α_r occurred. Moving an endpoint by $O(N^{-1})$ changes the integrals by $o(1)$ uniformly, by logarithmic integrability. There are at most N^2 pairs (i,j), contributing only $o(N)$. If $b = 0$, the finite sum has only $i = 0$, so only $\mu = 0$ is allowed. If $c = 0$, the left polynomial vanishes because $d - j \ge \lfloor N/2 \rfloor > 0$.

Proposition 6.1 (Exact continuum envelope)**.** *For $c > 0, b \ge 0$,*

$$\sup_{\substack{0\le\delta\le 1/2 \\ 0\le\mu\le\nu\le 1/2}} F_{\mathrm{L}}(\mu,\nu,\delta) = \max\{\log c, \mathcal{Q}(x,b,y) + \tfrac{1}{2}\log c\}. \tag{6.3}$$

For $b = 0$ restrict the left side to $\mu = 0$. This is an equality of continuum upper envelopes, not a lower bound for an oscillatory finite polynomial.

Proof. First assume $b > 0$, $0 < \delta < 1/2$, and set $z = \sin^2(\pi\delta)$, $\mathcal{D} = A^2 - 4Pz$. The ratio calculation in Lemma 5.2 shows that L_δ increases from zero to infinity before t_c and decreases from infinity to

$$L_{\mathrm{end}} = \frac{A + \sqrt{\mathcal{D}}}{2z}$$

afterward. The endpoint derivatives are

$$\partial_\mu F_{\mathrm{L}} = \log b - \log L_\delta(\mu), \qquad \partial_\nu F_{\mathrm{L}} = \log L_\delta(\nu) - \log c.$$

A stationary $\nu < t_c$ is a minimum, and a stationary $\mu > t_c$ is a minimum. The logarithmic singularity creates no maximum: the derivative in ν is positive on both sides near t_c, and that in μ is negative. The boundary $\mu = 0$ is excluded for $b > 0, \nu > 0$ by its positive initial derivative. Apart from the diagonal $\mu = \nu$, the possible maxima therefore have $\mu = \mu_b$ and either $\nu = 1/2$ or a stationary $\nu_c > t_c$.

The diagonal contributes at most $\max(\log c, \frac{1}{2}\log b + \frac{1}{2}\log c)$. The terminal value is $Q_b(\delta) + \frac{1}{2}\log c$, because the complete prefix integral is zero. Both are bounded by the proposed answer, using Lemma 5.1 and $\mathcal{Q}(x,b,y) \ge \frac{1}{2}\log b$.

A post-change stationary endpoint exists precisely when $c \ge L_{\mathrm{end}}$. If $c \le \max(x,y)$, this never occurs in the open interval. Otherwise it begins at δ_c, where

$$\sin^2(\pi\delta_c) = \frac{A}{c} - \frac{P}{c^2}, \qquad \sin^2(\pi\nu_c) = \frac{c^2\cos^2(\pi\delta)}{(c-x)(c-y)}. \tag{6.4}$$

At entry $\nu_c = 1/2$, and afterward $t_c < \nu_c < 1/2$. Define

$$J_c(\delta) = \nu_c \log c + \int_0^{\nu_c} \log|a_\delta(t)|\, dt + \int_{\nu_c}^{1/2} \log \lambda_\delta(t)\, dt.$$

The straddling value is exactly

$$S_{\mathrm{L}} = Q_b(\delta) + \log c - J_c(\delta). \tag{6.5}$$

Subtracting the displayed integrals and using (4.6) proves this with no additive constant.

Put $T_c^- = \sqrt{c^2 z - cA + P}$ and $\mathcal{H}_c = 2cz - A + 2\sqrt{z}\,T_c^-$. On the strict branch,

$$J_c' = -\log(\mathcal{H}_c/\sqrt{\mathcal{D}}). \tag{6.6}$$

Here is a normalization proof. The stationary envelope rule gives $\partial_{\log c} J_c = \nu_c$, while direct differentiation gives

$$\partial_\delta \nu_c = -\frac{c\sin(\pi\delta)}{T_c^-}, \qquad \partial_{\log c} \log \mathcal{H}_c = \frac{c\sin(\pi\delta)}{T_c^-}.$$

Thus the difference in (6.6) is independent of c for each δ. At $c \downarrow L_{\mathrm{end}}$, one has $\nu_c \to 1/2$, $T_c^- \to 0$, and $\mathcal{H}_c \to \sqrt{\mathcal{D}}$. The complete prefix integral has zero derivative. The remaining tail shrinks where $a_\delta(1/2) \neq 0$, so $J_c' \to 0$. This fixes the constant. To justify differentiation despite the interior logarithmic zero, write $\ell(t) = \log|2\cos(\pi t)|$. At fixed $\nu > t_c$, the prefix integral is

$$\int_\delta^{\nu+\delta} \ell(t)\,dt + \int_{-\delta}^{\nu-\delta} \ell(t)\,dt.$$

Its derivative is $\ell(\nu+\delta) - \ell(\nu-\delta)$; the two terms at $\pm\delta$ cancel. The moving zero lies strictly inside the integration interval, while all its endpoints are regular. Thus this primitive differentiates normally on the strict branch. At $\nu = 1/2$ its derivative is zero. The tail integral has smooth integrand there, so the stated branch-entry limit follows.

It follows that $S_{\mathrm{L}}' = \log(\mathcal{H}_c/\mathcal{E}_b)$. Both positive quantities are larger roots with the same product $\mathcal{D}$:

$$\mathcal{E}_b = C_b + \sqrt{C_b^2 - \mathcal{D}}, \quad C_b = A + 2bz,$$
$$\mathcal{H}_c = C_c + \sqrt{C_c^2 - \mathcal{D}}, \quad C_c = 2cz - A.$$

Both $C_b, C_c \geq \sqrt{\mathcal{D}}$. The larger-root function is increasing, and hence

$$\operatorname{sign} S_{\mathrm{L}}' = \operatorname{sign}((c-b)z - A).$$

As z increases, this sign can pass only from negative to positive. There is no interior spectral maximum. At branch entry the value is the terminal value already bounded. At $\delta \to 1/2$, both $\mu_b, \nu_c \to 0$, and logarithmic integrability gives $S_{\mathrm{L}} \to \log c$. The upper bound follows.

The continuum candidates are attained: $\mu = \nu = 0$ gives $\log c$, and $\delta = 0, \nu = 1/2, \mu = \mu_b$ gives $\mathcal{Q}(x, b, y) + \frac{1}{2}\log c$. For $b = 0$, use its sole start or the continuous limiting formulas. For $P = 0$, the exact one-step root gives the same identities without division by P. The exceptional $\mathcal{D} = 0$ occurs only at $x = y, \delta = 1/2$, where the computed continuous endpoint values apply. □

7 Face phases and the last spectral cap

To use the signed real transfer at arbitrary face phases, we need positivity of the interval continuant itself, rather than positivity of its recurrence coefficients.

Lemma 7.1 (Continuous real positivity of interval tails)**.** *For $0 \le i \le j \le m$ and $0 \le \delta \le 1/2 - 1/(2N)$, $K_{i,j}$ has nonnegative coefficients in x, y. Consequently*

$$|K_{i,j}(X_0, Y_0; \delta)| \le K_{i,j}(|X_0|, |Y_0|; \delta).$$

Proof. Set $U_r = K_{i,i+r} / \prod_{v=i+1}^{i+r} s_v^2$. Then $U_0 = 1, U_1 = x + y$, and for $r \ge 2$, $j = i + r$,

$$U_r = (x + y)U_{r-1} + \beta_j xy U_{r-2}, \qquad \beta_j = \frac{\alpha_{j-1}}{s_j s_{j-1}}.$$

The recurrence starts at its second step, so it never divides by s_0. Product-to-sum gives the row-independent identity

$$\alpha_{j-1} + s_j s_{j-1} = 4\cos^2(\pi\delta) - 4\sin^2(\theta/2) \ge 0.$$

Thus $1 + \beta_j \ge 0$, even if $\beta_j < 0$. Simultaneous induction using

$$U_r - xU_{r-1} = y(U_{r-1} - xU_{r-2}) + (1 + \beta_j)xyU_{r-2}$$

and $U_0 = 1, U_1 - xU_0 = y$ proves positivity. Taking moduli in a polynomial with nonnegative coefficients proves the last assertion. □

In (6.1), take absolute values of the middle monomials and of the scalar unused products. The lemma then permits the positive outer moduli in the *same signed* continuant. Thus Proposition 6.1 bounds every face phase uniformly for

$$-2 \le \sigma \le b_N := 2\cos(\pi/N).$$

No positivity of the full numerator, or comparison of its unused products to their endpoint values, has been asserted.

For the small remaining interval, let p be any complex polynomial of degree at most d, bounded by M on $[-2, b_N]$. The affine coordinate of $\sigma = 2$ relative to this interval is $z_N = (6 - b_N)/(b_N + 2)$. For a degree-$d$ polynomial r, the function $w^{-d} r((w + w^{-1})/2)$ is holomorphic outside the unit disk and at infinity. On its boundary it is bounded by $\|r\|_{[-1,1]}$. The maximum principle on the reciprocal disk gives

$$|r(z)| \le \|r\|_{[-1,1]} \exp(d \operatorname{arcosh} z), \qquad z \ge 1.$$

Complex coefficients cause no change. As $z_N = 1 + O(N^{-2})$, uniformly for $N \ge 3$,

$$\sup_{-2 \le \sigma \le 2} |p(\sigma)| \le M \exp(d \operatorname{arcosh} z_N) \le CM. \tag{7.1}$$

For each fixed face-phase tuple the actual left polynomial has spectral degree at most d. Apply (7.1) before taking the supremum of those phases. The constant is independent of them. This proves the left assertion of Theorem 1.1. Row reflection proves the right assertion. The central assertion was proved in Section 5. Every compact estimate respects the declared zero pattern, completing the theorem.

8 Use in a shared coefficient estimate

A fixed finite collection of boxes may be assigned genuine labels s_e, with every occurrence of each face variable identified. Keep designated zero faces exactly zero. Let t_f be the ordinary coefficient targets, and choose a single positive radius ρ_f for each remaining face. Require that the sum of the two outer moduli is positive in every box; a zero pattern killing both outer variables is outside Theorem 1.1. Cauchy's coefficient formula gives

$$\left|[L^t]\prod_e H_{V,s_e}(L;\sigma_e)\right| \le \prod_f \rho_f^{-t_f} \sup_{|L_f|=\rho_f} \prod_e |H_{V,s_e}(L;\sigma_e)|.$$

Theorem 1.1 applies at this shared radius vector. If $t_f/N \to \tau_f$, the uniform real-cube upper rate is at most

$$\sum_e A_{s_e}(\rho) - \sum_f \tau_f \log \rho_f.$$

The number of boxes is fixed before N grows, so their errors sum to $o(N)$. For finitely many sector strings, take the maximum of these sums before bounding their finite sum. This keeps correlations between genuine labels; no Cartesian enlargement is needed.

For an actual matched spherical diagram, the exact marked coefficient and signed-power filling identities of [1, Theorem 9.1 and Corollary 9.2] identify the shared polynomial with its original tensor source. Its Theorem 1.1 then bounds the cost of the finite Hermite functionals. These are separate uses of an exact source identity and an operator norm. They do not supply a strict source gap at arbitrary radii. Nor does this paper produce a nonzero leading contribution or identify its rate with a hyperbolic volume.

The central coefficientwise comparison (3.3) has not been extended to the one-sided polynomials. Their theorem is a weighted value bound after the signed recurrence, both unused products, and the final real spectral cap have all been accounted for.

References

[1] *Finite spectral evaluation and uniform filling bounds*, Sections 2–5 and 9–10. Companion paper.